

Five Readers on *The Turn*

A referee exercise on *Finding Mind*, draft 9

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with a synthesised revision plan

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How to read this

Five reviewers, each reading the same 570-page draft with different commitments, each finding different soft spots. The voices are simulated; the criticism is not. Where a reviewer speaks in the first person, treat the idiom as a device for keeping the commitments distinct and the objection sharp — nothing here should be attributed to the actual person named.

Scope is *The Turn* only: Part I (Machinery, ch. 5–17), Part II (Ecology as Cognitive Architecture, ch. 18–21), Part III (Situating the Learner, ch. 22–30), Part IV (Why Scientists Get Misled, ch. 31–40), Part V (Entropy, Choice, and the Break from Turing, ch. 41–52), Part VI (Origins of Life, ch. 53–61), and the Conclusion. The Pledge and The Prestige are referred to only where The Turn depends on them.

Locations are given by source file and label, since you asked for the L^AT_EX source. The build tree used was `draft2` → `...` → `draft9` overlaid onto `ThePledge/`, `TheTurn/`, `ThePrestige/`, with `TheTurn/turn_causality_ch02.tex` deleted, per `INTEGRATION-NOTES-7.md`.

Five findings recur across reviewers and are collected in §7 so they are not lost in the individual reports. Three are internal contradictions the draft can repair without new research. One is a broken proof. One is a piece of apparatus that was retracted in one chapter and is still load-bearing in two others.

Contents

How to read this	1
1 David Spivak reads the machinery	3
2 Curt Jaimungal reads for the crux	6
3 Ben Goertzel reads for the architecture	8
4 Max Tegmark reads the physics	11
5 Joscha Bach reads for the subject	16
6 Where the five disagree	20
7 Cross-cutting findings	22
8 Revision plan	24

1 David Spivak reads the machinery

The commitments

Applied category theory. If a construction is called a functor, a monad, a fibration, or an adjunction, the coherence data should be exhibited or the word should be withdrawn. Universal properties are what make a categorical framing earn its keep; without them the diagrams are decoration.

What works, and why I am reading the rest carefully

Chapter 9, *The Category of GSLTs*, is the best piece of category theory in the book and I want to say so before I complain. The three defects of the naive definition — the constant map, the fact that Milner’s encoding is not a map of terms, the ambient-context problem — are diagnosed correctly, and taking contexts as primary fixes all three at once for a reason, not by fiat. Defining morphisms as pseudofunctors of context multicategories that are bisimulation-preserving *with the target bisimulation computed over the image of the source’s contexts* is exactly right. That last clause is the whole game and most people writing about expressiveness get it wrong.

So I will hold the rest of the book to that standard.

Φ is not a fibration

`turn_hypercomp_ch02.tex, sec:fibration:`

$$\Phi : \mathbf{GSLT}_{\mathbf{TC}} \times \mathbf{Ord} \longrightarrow \mathbf{GSLT}.$$

This is a functor *out of* a product. A Grothendieck fibration is a functor $p : \mathcal{E} \rightarrow \mathcal{B}$ with cartesian lifts. What you have written down is an indexed family — possibly a pseudofunctor $\mathbf{Ord}^{\mathbf{op}} \rightarrow \mathbf{Cat}$, if you pick a direction — and the Grothendieck construction would produce a fibration from it. It is not itself one.

Worse, the direction is overdetermined. You give both $\pi_\alpha : \Phi_{\alpha+1} \rightarrow \Phi_\alpha$ and $\iota_\alpha : \Phi_\alpha \rightarrow \Phi_{\alpha+1}$, and $\mathbf{Ord}$ is a poset, so a functor out of it can only be one of these. You have described an adjoint pair or a section–retraction, and neither is named. If $\iota_\alpha \dashv \pi_\alpha$ — which I suspect is what you want, since inclusion adds and projection forgets — then say so and check the triangle identities. The whole tower would then be a tower of adjunctions, which is a much better object than what is currently on the page and costs you nothing you do not already have.

The book contains two objects called “the fibration” and they are not the same object. `turn_hypercomp_ch02.tex` asserts Φ over $\mathbf{GSLT}_{\mathbf{TC}} \times \mathbf{Ord}$. `turn_ndo_ch02.tex` open problem (ii) asks someone to “exhibit the fibration $p : \mathbf{Hyp} \rightarrow \mathbf{GSLT}$ rigorously and test whether angelic and demonic resolution are the Lawvere adjoints $\exists \dashv (-)^* \dashv \forall$ ”

to reindexing.” The second is the right object and is flagged as unbuilt; the first is the wrong object and is asserted as built. A reader who takes the chapters in order meets the assertion first.

The Lawvere-adjoint question in that open problem is the correct one and I would push harder on it than the draft does. Angelic and demonic resolution are $\exists$ and $\forall$ over the fibres of the pairing family; if the fibres have the requisite limits and colimits you get the adjoint triple and with it a Beck–Chevalley condition, which is precisely the coherence statement that “choice strength is stable under reindexing” — which is *the same question* as Question `ndo:q:crux`, the stability of rigidity under descent, and the same question again as subject reduction in `turn_chs_ch01.tex`. The book says twice that two of these are the same question. It is three, and the third framing is the one that tells you what to prove.

Three monads, three adjunctions, one missing distributive law

`turn_gslt_ch06.tex` gives `Cost`, `turn_gslt_ch07.tex` gives `Hist`, `turn_gslt_ch10.tex` gives $\mathbb{K}$, and each is resolved by a free/forgetful adjunction. Good. The unit and multiplication for `Cost` are given explicitly and the laws are derived from the two constituent monoids, which is the right way to do it. The non-idempotence remark is exactly the observation a resource monad deserves.

But the book applies these monads *together* everywhere — Part III decorates a cost-accounted theory that also has a history, Part V meters a logged theory — and nowhere is a distributive law $\text{Cost} \circ \text{Hist} \Rightarrow \text{Hist} \circ \text{Cost}$ (or the reverse) stated, let alone verified. Without one, “a cost-accounted theory with recorded histories” is not a well-defined object; it is two constructions applied in an order nobody has fixed.

And you have the material to know this matters. `turn_entropy_ch01.tex`, `rmk:two-erasures`, says the two counits do not factor through each other and that “the failure to factor is graded — there is a lattice of intermediate laxities.” That remark is, read categorically, the observation that no distributive law exists in either direction and that what you have instead is a family of lax interchange maps. **That is a genuinely interesting mathematical situation and the book treats it as an aside in a chapter about entropy.** It belongs in Part I, stated as a proposition, with the lattice of laxities exhibited. It would also make the entropy chapter’s central move — that the gap between the erasures is “a supply of positions” — rest on something rather than on a metaphor.

The compactness proof is wrong

`turn_sciencebot_ch06.tex`, `thm:compact-consensus`. You define φ_n so that it is satisfied by agents that have completed *at least* n oscillation steps. Then $\llbracket \varphi_1 \rrbracket \supseteq \llbracket \varphi_2 \rrbracket \supseteq \dots$, so $\bigcup_{n \leq N} \llbracket \varphi_n \rrbracket = \llbracket \varphi_1 \rrbracket$, and an agent that has oscillated more than N times is *in* that union. The claim that it is uncovered is false, so there is no contradiction with compactness and the theorem is unproved.

The intended argument is presumably the other way round: take the complements, or index so that φ_n says “has completed *fewer than* n steps,” giving an increasing open cover of the non-oscillating agents whose finite subcover bounds the depth. That version may well work. As written it does not, and this is the theorem the Conclusion singles out as “the sharpest result of that part” (`turn_conclusion_ch01.tex`).

Scott or Stone? The book cannot have both

Same chapter, `def:log-top`. You generate the topology from HML extensions as a subbasis and then say “this topology is the standard *Scott topology*... sober, T_0 , and quasi-compact.”

If your HML has negation — and `turn_causality_ch05.tex` and the namespace logic of `ch:sci` certainly do — then every subbasic open is also closed, the space is Hausdorff and totally disconnected, and you are in *Stone* duality, not Scott. Points are closed, the specialisation order is discrete, and “sober, T_0 ” understates it badly. If you drop negation to get a genuine Scott topology, you lose the ability to say $\neg\varphi$, which the consensus theorem quantifies over.

This is not pedantry about which duality to cite. The two settings give different answers to “what does compactness of a population mean?” In the Stone setting compactness is a statement about a profinite limit and connects directly to the ultrametric structure that `ch:sci §sci:sec:lattice` already uses for the hypothesis space — which would unify two parts of the book that currently do not speak. **Decide which topology, and the sciencebot part gets shorter and better.**

Smaller things, in the order I met them

- `turn_gs1t_ch05.tex` defines *hosting* as “faithful” and *exhausting* as “dense” in parentheses, then never uses the categorical notions again. If they are faithfulness and density of a (pseudo)functor, say so precisely; if they are analogies, drop the parentheses.
- `prop:assembly-index` (`turn_origins_ch02.tex`) is labelled a proposition and has no proof. It is a definition — or, if it is a claim about Assembly Theory’s assembly index, it needs a translation functor and a preservation theorem. See Tegmark, §4, who is angrier about this than I am.
- `turn_gs1t_ch10.tex`’s monad proposition asserts a free/forgetful adjunction between $\mathcal{L}$ and $\mathcal{L}_\Sigma$ and gives the action on morphisms by transport, but does not say what the free extension is free *on*. For Cost you did say. Say it here.
- Part I promises “three constructions, three monads, three free/forgetful resolutions” and delivers three propositions of quite different strength: Cost is worked, Hist is sketched in four sections and 546 words, and $\mathbb{K}$ is asserted. A status column of the kind `turn_causality_ch10.tex` uses so well would help here too.

2 Curt Jaimungal reads for the crux

The commitments

Where is the load-bearing claim, stated so a smart non-specialist can hold it and then argue with it? How does it differ from the four other frameworks the audience already knows? And — the question I ask everyone — what would have to be true for you to abandon it?

You have three theses and the book never picks one

Reading *The Turn* end to end, I can find three candidate load-bearing claims, each stated as though it were the point.

1. *Mind is bounded by manufacture.* `turn_origins_ch08.tex`, `thm:ai-bound`, and the Conclusion’s closing paragraphs. An ecology assembled by n joining steps resolves its world at best to n -step bisimulation.
2. *Consciousness is a coordinate, not a threshold.* `turn_ndo_ch02.tex` opening, `turn_hypercomp_ch05.tex`. Whether every agent has a position in the Weihrauch lattice or crosses into a stage.
3. *The noumenal survives as a budget constraint.* `prestige_ch01.tex` §`sec:kant-affordable` — and you have said elsewhere this is the pivot of the whole book.

The third is the one that would make an audience sit forward, and it is in *The Prestige*. **The Turn builds it, states it in passing in `turn_scope_ch01.tex` §“What this is for,” and hands the payoff to a section the reader reaches two hundred pages later.** A reader who stops at the end of *The Turn* — and many will, because *The Turn* is where the mathematics is — never learns what the framework was for.

The Overarching Introduction opens on “what is the minimum mathematical structure needed to ground physics, knowledge, life, and mind?” That is a scope statement, not a thesis. Scope statements promise; theses commit.

The comparison the book owes and does not pay

Three sentences in `turn_hypercomp_ch08.tex` handle Friston and Tononi, and that chapter is visibly stale — it still describes the material as “the three papers in this series” and calls the ordinal treatment of consciousness “novel” after two later chapters have argued the ordinals are wrong. Nothing anywhere engages Wolfram’s ruliad, Levin’s work on scale-free cognition, global workspace theories, or — most conspicuously — Assembly Theory’s own critics, in a part that spends a hundred pages engaging Assembly Theory.

What the book needs is one page, early, in this shape:

Framework	Its measure	Where this book disagrees
IIT	Φ , integration over a partition	Consciousness is not integration but affordable resolution; the partition problem becomes the individuation fixed point of ch:eng
FEP / active inference	Free-energy minimisation	Prediction error is priced in a conserved currency the agent can run out of; there is no scorekeeper
Assembly Theory	AI and CN	Both are supplied a region and an observer; the two stop being independent
Ruliad / Wolfram	Computational irreducibility, observer coarseness	The coarseness has a budget, and the budget is a physical quantity
Global workspace	Broadcast and access	Access is metered; eng:prop:fragile prices the broadcast at p^d

Every row of that table is already argued somewhere in The Turn. None of them is collected.

“Precisely enough to be wrong” appears four times

The phrase is in the Turn frontispiece, in the Overarching Introduction, and twice more including the last line of the Conclusion. Repetition of a defensive claim reads as anxiety, and the book does not need it: [turn_causality_ch10.tex](#)’s CON/THM/CNJ status column does the work honestly and is the single best editorial decision in the draft. **Say it once, and extend the status column to every part.**

Right now the reader can audit Part III’s claims because that part has the column, and cannot audit Part V’s or Part VI’s because they do not. Part V is where the boldest claims live.

The words *sentience* and *consciousness*

[turn_hypercomp_ch04.tex](#) defines sentience as a scalar running integral of predictive accuracy that gates the budget. That is not what anyone means by sentience; it is closer to what an economist would call solvency and what a psychologist would call, at most, a mood-congruent cognitive-resource effect. [ch05](#) defines consciousness as oracle access, which is a claim about computational power.

You are entitled to appropriate the words. But an appropriation costs a page of defence and currently costs a sentence. Either write the defence — here is what ordinary usage tracks, here is why the technical notion deserves the name, here is what is lost — or rename. “Affect budget” and “resolution coordinate” would cost the book nothing except the frisson of the titles.

The entry problem

570 pages, and the best material — Part II, the ecology — starts at p. 141 behind seventy pages of category theory. The Overarching Introduction tells the reader they may begin at Part II, but nothing structural supports the advice: Part II’s four chapters open with bridges that reach back into Part I’s notation, and the reader has no signal about which Part I constructions they will actually need.

Two cheap fixes. Give Part II a two-page “what you need from Part I” opening that states the four or five constructions it uses and where they are. And move one worked thing forward — the noughts-and-crosses world of `turn_rholife_ch02.tex` is concrete, has numbers, and would function as an overture if a compressed version of it opened The Turn instead of the machinery.

What would change your mind?

The book answers this well in exactly one place and badly everywhere else. `turn_entropy_ch01.tex` §“What this buys” says: if every aperture in a physically realisable system carries only finite contention, the framework predicts no physical system exceeds the Turing computable, “and the question would be settled the other way.” That is a real falsification condition and it is stated in a subordinate clause. **It should be a section, because as Tegmark will point out, the antecedent looks true.**

Elsewhere: what observation would refute “mind is bounded by manufacture”? What would refute the identification of assembly index with minimum path length? The book’s own answer — that these are constructions, true by definition of the objects — is honest, and it means the falsifiable content is concentrated in Part VI’s three empirical remarks (`scm:rmk:biosig`, `scm:rmk:shed`, `scm:rmk:groupsize`). Those are worth a chapter of their own, and they are currently remarks.

3 Ben Goertzel reads for the architecture

The commitments

Does this build anything? An account of mind that cannot be turned into a running system is a philosophy of mind, which is fine, but this book keeps promising an architecture. Show me the learning algorithm, the inference control, the credit assignment, and the path to open-ended novelty.

The confinement result is the best thing here and it is half-used

`sci:prop:confinement` — a walk whose steps are all shorter than 2^{-n} never leaves the ball of that radius — combined with `sci:rem:nogradient` — the hypothesis space is an ultrametric tree and admits no gradient — is a genuinely important negative result about incremental

learning, and I do not think it is widely appreciated outside this book. Gradient-free, ultra-metric, provably confined: that is a precise statement of why hill-climbing in program space does not work, and it is derived rather than observed.

The book then draws the right conclusion — the unit of learning has to move — and hands the escape entirely to recombination. `turn_rholife_ch01.tex` §`sci:sec:repro` does this in about two thousand words, and the crossover mechanism (the germ is a quotation, the normal form is a tree, a locus is a context, crossover swaps subtrees at loci) is the single most exciting construction in the book.

It should be a chapter. Possibly a part. Genetic programming has thirty years of results on exactly this operation — destructive crossover rates, bloat, schema theory, the ineffectiveness of subtree crossover on typed program trees — and none of it is cited. More importantly, your version has something GP does not: the loci are *contexts* in a calculus with reflection, so the type discipline that GP lacks is available for free from the OSLF construction. That is a publishable result on its own, and it is currently a subsection.

Where is the inference?

§`sci:sec:search` gives the geometry of the hypothesis space, the constraints any revision policy must satisfy, and “a catalogue of strategies,” and then explicitly declines to fix a policy (§`sci:S1.4`: “We are *not* claiming to have fixed a policy, proved a convergence theorem, or said anything about statistical efficiency”). For a research note, defensible. For a book chapter in a part called *Ecology as Cognitive Architecture*, it is the hole where the architecture goes.

You have the pieces for a real answer and they are scattered:

- The relaxation lattice (§`sci:S5`) is an inference structure. Generalisation forgets a conjunct, specialisation adds one — that is version-space learning, and version spaces have known complexity results.
- The essay is a guarded receipt, so hypothesis evaluation is *reduction*, not interpretation. That means inference control is scheduling, and you have an entire part (§`ch:apr`, §`ch:agy`, §`ch:chs`) about the strength of schedulers. Nobody has connected them. **Inference control in this framework is a choice-type discipline.** That is a one-paragraph observation that would tie Part II to Part V and neither part makes it.
- Credit assignment is the foraging inequality (§`sci:sec:game`): a hypothesis that says where the tokens are is paid in tokens. That is the answer to the scorekeeper problem and it is genuinely elegant. But it gives you credit assignment only for hypotheses that were *acted on*. What about the counterfactual — the hypothesis that would have paid?

No relationship to MeTTa or Hyperon, and you built MeTTaIL

MeTTa appears once in the whole of *The Turn*, in a table in `turn_causality_ch10.tex`. Given that the same research programme produced a MeTTa language-definition platform, and

given that Hyperon’s Atomspace is a metagraph with rewriting and Distributed Atomspace pattern matching, the absence is conspicuous to anyone who knows both bodies of work.

Two things I would want said explicitly. First: is a MeTTa program a GSLT in your sense, and if so what does the OSLF construction emit for it? That is a concrete, checkable question and answering it would give the framework a second worked instance besides rho, which it badly needs — right now every construction is illustrated on rho and a reader cannot tell which results are about GSLTs and which are about rho. Second: Atomspace is a shared namespace with concurrent readers and writers, which is exactly your `ch:com` setting. Your composition-of-learners results — disjointness is factorisation, overlap is graded, non-interference is not stable — are results about Atomspace-style architectures and nobody in that community knows it.

The assembly-index bound is a bound on a closed system

`thm:ai-bound` says an ecology of assembly index n resolves its world at best to n -step bisimulation, and `rmk:power-caveat` carefully says this bounds resolution and not reach. Good. But the slogan — “Mind is bounded by manufacture” — overstates the theorem it summarises, in a specific way the book should address.

Minds build instruments. A microscope is manufacture that raises the resolution of the thing looking through it without raising the assembly index of the looker. In your terms: an ecology can construct a term whose interaction with the environment yields distinctions the ecology could not make unaided, and then read the result at low depth. That is what every scientific instrument is.

So $AI(E)$ bounds what E resolves *by itself*, and minds are distinguished precisely by not doing things by themselves. The repair is not hard — the bound presumably applies to E together with its instruments, which is a larger ecology with a larger index — but it has to be said, because as stated the theorem’s most natural reading is false and the slogan makes the false reading the memorable one.

Open-endedness

Nothing in the framework generates genuinely new hypotheses. The hypothesis language is emitted once by the OSLF construction from a fixed presentation, and the learner then searches a lattice that was determined before it was born. `sci:S1.3` presents this as a virtue: the language is “generated, not designed,” adequate for context-labelled bisimulation and no more.

But a mind that can extend its own presentation — adding a sort, a rule, an equation — is a different and more interesting object, and your machinery does not obviously forbid it. Reflection means a term can quote code; the germ mechanism means quoted code is heritable; `ch:eng`’s engines mean new token colours can appear. **What would it take for a learner to extend the GSLT it lives in?** If the answer is “nothing, this is already available,” say so, because it is the difference between a framework for bounded learners and a framework for

open-ended ones. If the answer is “the framework forbids it,” that is a substantive limitation and should be in the open problems.

One small thing that bothered me

`scm:rmk:oneg`: a mind with k skills does not pay the germ level k times, because “the germ quotes the whole individual’s code, and quotation does not distribute over the parts.” That is right, and it is also the argument for why modular acquisition of new skills should be cheap in this framework — which is a claim about transfer learning and lifelong learning that nobody makes. `turn_rholife_ch01.tex` §“Why continual learning falls to the ecology” is nearly there. Connect them.

4 Max Tegmark reads the physics

The commitments

Numbers, units, and a prediction that could come out wrong. A correspondence between a mathematical structure and a physical one is worth having only if it constrains something. And when a book says four times that its claims are “stated precisely enough to be wrong,” I want to know which one I should try to falsify first.

Start with the thing I would try to falsify, because you have it and you buried it

`turn_origins_ch09.tex`, `scm:prop:geom` and `scm:rmk:falsepos`. An instrument whose resolution falls short of a population’s separation depth by k does not miss copies — it reports b^k times too many. And the inflation is largest exactly where the population is most diverse, which is exactly where selection has been most active. So the error is not noise; it is a bias aligned with the signal.

This is the best result in draft 9 and possibly in the book, and it is a remark inside a subsection. It is a concrete, quantitative, practically consequential claim about the leading experimental biosignature programme, and it is stated in a form a mass spectrometrists could act on. Cronin’s group measures CN by counting identical fragmentation spectra; two molecules with indistinguishable MS2 spectra are counted as copies. That is precisely a finite-resolution equivalence, and your proposition says it inflates the count exponentially in the shortfall.

The book then says “we do not know how large the effect is in mass-spectrometric practice and we would want it estimated.” You can estimate it. A tandem MS instrument has a known number of distinguishable fragmentation channels; a molecular population has a known isomer diversity for a given formula. Even an order-of-magnitude Fermi estimate — is k typically 1, or 5? — turns this from a philosophical caution into a number somebody has to answer. Do the estimate, get it wrong if necessary, and put it in the abstract of whatever paper this becomes.

10^{80} agents

`turn_sciencebot_ch05.tex`: a population of $|\mathcal{P}_0| \approx 10^{80}$, “comparable to the number of quarks in the observable universe,” with the remark that the scale is chosen so that “every cluster of agents is itself a large enough system to implement a complex agent.”

Two problems, and the second is the serious one.

First, 10^{80} is the baryon count; the quark count is about three times that, and both are dwarfed by the photon count ($\sim 10^{89}$). Minor, but it is the kind of thing that makes a physicist stop trusting the rest.

Second, and fatally for the framing: if there are 10^{80} agents in the observable universe, each agent has on average the mass–energy of one proton. A proton cannot implement a complex agent. So the remark justifying the scale — that clusters at every level are large enough to be agents — is inconsistent with the scale it justifies. You cannot have 10^{80} agents and have the agents be complicated.

The repair costs nothing, because you do not need the number. What the argument needs is: *a population large enough that the communication diameter exceeds the coherence length*, whatever those are. State it that way, scale-free, and the result gets stronger rather than weaker — it now applies to a bacterial colony, a research community, and a galaxy-spanning civilisation alike, which is presumably what you meant. Right now the physical gloss is doing nothing but attracting fire.

The retracted apparatus is still load-bearing in two places

`turn_causality_ch09.tex` is admirable. It states the complex decoration, states the obstruction — bisimulation is branching-time and refuses to pool runs, so no assignment of weights makes paths annihilate — explicitly retracts the density-matrix proposal, and names three routes out without pretending to have chosen. That is how to withdraw a claim.

The problem is that two later chapters never got the memo.

`turn_sciencebot_ch05.tex` **defines message fidelity as $|W(\gamma)|^2$, the squared modulus of the path amplitude**. The whole of Part IV — communication radius, degradation at scale, coherence clusters, the nerve, the apparent ontology — is built on that definition. It is the retracted apparatus.

`turn_hypercomp_ch05.tex` **defines ϵ -approximate oracle access via $|\langle \text{halt} \mid A \otimes [(M, x)] \rangle|^2$** , “where probability is measured by the path integral.” Same apparatus. And `sec:hyper-gaps` Gap 7 compounds it by asking for the oracle interaction amplitude to be made precise “via the quantum Gillespie algorithm of Part I” — which `turn_causality_ch09.tex` declines to call a quantum dynamics.

Neither use actually needs amplitudes. Part IV needs a fidelity that degrades multiplicatively along a path and is bounded below by a threshold; the stochastic instance ($S = \mathbb{R}_{\geq 0}$), which the book says is unobstructed and is what the implementation runs, supplies exactly

that. Rewriting `turn_sciencebot_ch05.tex` in the stochastic semiring is a half-day of work and removes the book’s largest internal inconsistency. Part V’s graded consciousness needs a probability of correct answer, which is the same fix.

“Least action is the tropical path integral”

`turn_causality_ch06.tex`, `prop:leastaction`. I want to be careful here because the chapter is a substantial improvement on what it replaced — the honest paragraph explaining that $S = \sum \log \text{dec}$ was “a change of variable, not a variational principle” is exactly right and rare.

But the replacement oversells too, in a different direction. In the tropical semiring $\oplus = \min$, so the transition weight is the minimum over paths, and the proof is one line: “immediate from $\oplus = \min$ and $\otimes = +$.” That is true and it is a restatement of the definition of the tropical semiring. What it gives you is *Bellman’s principle* — the shortest-path structure of dynamic programming — which is a real and useful thing, but it is optimal control, not Lagrangian mechanics.

The classical principle is $\delta S = 0$: the realised trajectory is a *stationary* point of the action over a continuum of nearby trajectories, and most physically realised trajectories are saddle points, not minima. Your construction has no continuum, no variation, no Euler–Lagrange equation, and no $\hbar$. Calling $\min_\gamma$ “the principle of least action” invites a physicist to look for those four things and conclude you do not know they are missing.

The honest and still-interesting statement: *in the tropical semiring the path integral computes the Bellman value function, and the classical variational principle would be recovered in a continuum limit that has not been constructed*. Then put the continuum limit in the open problems, where the Legendre transform already sits, correctly withdrawn.

Landauer, without a temperature

`turn_causality_ch07.tex`: the erasure a computation performs is bounded below by the holonomy of the energy form, and “the two thresholds coincide.” This is presented as the most satisfying result in the chapter and I understand why — structurally it is lovely.

But Landauer’s bound is $k_B T \ln 2$ per bit. It has units. It has a temperature. Your holonomy θ is a dimensionless accumulation around a cycle in a conversion graph. Saying the two thresholds coincide is, as far as I can tell, the claim that two abstract lower bounds happen to be the same abstract quantity — which is a theorem about your formalism, not a derivation of Landauer’s principle.

To make it a physical claim you need a temperature, and the framework has an obvious candidate it has not used: the free-energy structure of `turn_causality_ch07.tex` §“The second law” already distinguishes total value from available value, and the ratio of dissipation to entropy production *is* a temperature in every other thermodynamic formalism. **Define T in the framework, recover $k_B T \ln 2$ with the right constant, and this becomes a result rather than a resonance.** If the constant does not come out, that is worth knowing too.

Choice: two different lattices, one word

This is my most serious technical objection and it goes to the thesis of Part V.

`turn_entropy_ch01.tex` §“Choice is what fills the gap” argues: picking a winner from a contending family is a choice function; asserting a complete schedule exists is an instance of the axiom of choice; “finite contention is free, contention indexed by ω requires countable or dependent choice, class-sized families require the full axiom.” `turn_ndo_ch01.tex` §“A lattice, not a tower” then locates choice strength in the *Weihrauch* lattice of Brattka–Gherardi–Pauly.

These are different structures measuring different things.

- AC-strength is set-theoretic: which fragment of choice you must add to ZF. It is graded by cardinality of the index family.
- Weihrauch degree is computability-theoretic: the uniform reducibility of partial multivalued functions on represented spaces. LLPO, which the chapter cites as the home of demonic binary choice, is not an instance of AC at all — it is a semi-classical logical principle, and its Weihrauch degree concerns the uniform computability of a selection, not the existence of a set.

The two do interact — constructive reverse mathematics is exactly the study of that interaction — but the interaction is a research area, not a substitution. **Part V slides between them and the slide is load-bearing**, because the AC framing gives you the transfinite grading that motivates the ordinal tower, and the Weihrauch framing gives you the lattice that supersedes it. One paragraph naming both structures and stating which one the book uses — and what is being conjectured about the relation between them — would cost half a page and would convert an equivocation into an open problem.

And on the book’s own terms, finite contention settles it

`turn_entropy_ch01.tex`, last section: “if it turned out that every aperture in a physically realisable system carried only finite contention, then the framework would predict that no physical system exceeds the Turing computable, and the question would be settled the other way.”

Every physically realisable system carries only finite contention. There are finitely many degrees of freedom in a bounded region with bounded energy — that is the holographic bound, and it is about as well established as anything in this neighbourhood. So the antecedent holds and the book’s own falsification condition fires.

I do not think this destroys Part V. The interesting content survives in a different form: the *location* claim — that whatever excess capacity exists is spent at apertures, in dissipated free energy — is a claim about where computational resources go, and it is true and useful whether or not the resource is hypercomputational. What does not survive is the framing that the ordinal tower is inhabited.

So confront it. Make §“What this buys” into a chapter that states the finiteness objection in its strongest form — Bekenstein, holographic bound, finite contention — and then says

what is left. What is left is: choice strength is a resource that is spent, its grading is real and measurable in the Weihrauch lattice even at finite degrees, and the tower is a mathematical idealisation whose inhabitants are hypothesised. The book already says the last part (`turn_entropy_ch01.tex`: “the tower... remains a construction whose inhabitants are hypothesised rather than built”). Say it *first*, not last, and Part V becomes honest instead of looking like it is hoping the reader will not notice.

The worked composite: arithmetic sound, accounting slippery

I checked `scm:sec:composite`. The numbers are internally consistent — $\lceil 20 \cdot 0.95^{-3} \rceil = 24$, $\lceil 20 \cdot 0.90^{-3} \rceil = 28$, the copy series (1152, 48, 12, 1), the flat collapse 1213, the ratios 24 : 4 : 12, the 34.9% and 57.1%. All correct. Two accounting problems.

1. **Table `scm:tab:skills` carries two different depths in adjacent columns and the floor adds the other one.** The column d^* (optimal *integration* depth, from `eng:prop:fragile`) reads 3, 5, 3, 5; the column “separates at” (modal depth at which the skill’s learners become distinguishable) reads 4, 2, 3, 3. The assembly floor charges “differentia: d^* per skill, $3 + 5 + 3 + 5 = 16$ ” and cites L4 and L5 — but L4’s floor is $\text{AI} \geq n$ for *resolving* at depth n , which is the second column, summing to 12. If the identification is intended, one sentence should say why an integration depth and a separation depth are the same charge. If it is not, the floor is 24, not 28, and every percentage in the paragraph moves.
2. `scm:prop:learner` — “a learner costs strictly more than a scorer,” $\text{AI}(L) \geq \text{AI}(S) + 1 + n^-$ — is described as “the one bound here we find genuinely interesting,” and its n^- term never appears in the only worked evaluation. Either it is subsumed in the d^* charge, in which case say so, or the floor is missing a term.

Two places to cut

`scm:rmk:groupsize` derives a critical pod size of 75 from $\theta = 0.2$, $\rho = 5$, an assumed isoperimetric ratio of 0.3, and notes that “the literature on cohesive group size in social mammals lives in this neighbourhood.” Then: “we would not want that repeated as a result.”

If you would not want it repeated, do not print it. A number in the Dunbar neighbourhood, derived from four invented parameters, with a disclaimer, is exactly the artefact that gets screenshotted without the disclaimer. The *shape* of the claim — there is a finite closure size, it scales as $\rho/(\epsilon\theta)$, and it is set by the reliability of media *between* individuals rather than by anything about the individuals — is genuinely interesting and stands without the numeral. Keep the scaling law, delete the 75.

Similarly `turn_hypercomp_ch08.tex`’s “Broader Picture” section should go entirely. It describes the material as three papers in a series, credits the ordinal treatment as novel after the book has argued it is wrong, and disposes of IIT in a sentence that says the present approach “avoids the measure-theoretic machinery of IIT” — which reads as a boast about having less mathematics.

5 Joscha Bach reads for the subject

The commitments

A mind is a self-organising software process that constructs a model of its world and, crucially, a model of itself inside that world. The interesting question about consciousness is not why there is something it is like — that framing smuggles in the answer — but why a system would *represent itself* as having phenomenal states, and what work that representation does. Show me the self-model, the attention mechanism, and the motivational system, or you have described a very sophisticated thermostat colony.

You built reflection and used it for reproduction

This is my central criticism and everything else follows from it.

The rho calculus is reflective. A term can quote itself; quotation is inert; a name codes a strictly smaller term. `turn_rholife_ch01.tex §sci:S12.2` uses this beautifully: the germ is a quotation of the learner’s own code, transmissible and inert until dropped, and `scm:rmk:germgen` later identifies the germ with the generator of the individual’s scope. That is a genuine insight and it is doing inheritance work.

It is also, and the book never says this, a self-model. A learner that holds a quotation of its own code holds a representation of itself, in the same language it uses for hypotheses about everything else, in a form it can inspect without executing. The hypothesis language is a logic of namespaces; the learner’s own namespace is a namespace; therefore the learner can *form hypotheses about itself*, priced by exactly the same assay machinery.

Nothing in The Turn does this. Search the ecology part for a hypothesis whose subject is the learner: there is not one. The learner reasons about its environment, forages, composes, reproduces, and never once asks what it is. And yet:

- `sci:sec:exposure` — the exposure lemma — already prices *being known*. That is the third-person view of the same quantity.
- `ch:eng §eng:S8` defines an individual as a fixed point of a partition and says explicitly that the fixed point has multiple solutions. A system that must *choose* a solution to “what am I” is a system with a self-model, and the book calls this a circularity to be lived with rather than a cognitive function to be performed.
- `prestige_ch01.tex §sec:kant-affordable` says a learner “does not fail to see the distinctions it cannot afford — it fails to see that they are there, and reports a world with fewer kinds in it than the world has.” Apply that sentence to the learner’s model *of itself* and you have derived the structure of introspective illusion. The self-model is necessarily coarser than the self, the coarsening is not represented, and the system therefore reports a self that is simpler, more unified, and more transparent than it is.

That is an account of why a system represents itself as having phenomenal states, it is three

sentences long, it follows from machinery the book has already built, and it is not in the book. It belongs in The Turn, not The Prestige, because it is the payoff of Part II and Part VI rather than a metaphysical coda.

It would also let you retire Gap 8 ([sec:hyper-gaps](#): “the hard problem... remains open and is not claimed to be resolved here”), which is currently an abdication dressed as modesty. You do not have to solve the hard problem. You have to say why a budgeted reflective learner would generate the report that produces it. You can.

Consciousness as oracle access does not do what the book needs it to do

[turn_hypercomp_ch05.tex](#), [def:consciousness](#): an agent is conscious at level α if it has effective access to the oracle of stage α and its world model reflects the finer bisimulation quotient.

That is a definition of *being cleverer*. A philosophical zombie with a halting oracle is a zombie with a halting oracle. Nothing in the definition makes the agent’s world *appear* to it, makes the distinctions available to it *as distinctions it is making*, or distinguishes an agent that resolves finely from an instrument that resolves finely.

The remark that follows — “the phenomenological richness of higher consciousness is a direct consequence of ontological richness” — asserts the bridge rather than building it. And the book knows better elsewhere: [turn_sciencebot_ch08.tex](#), *The Apparent Ontology*, is entirely about the gap between what is there and what appears, and it is the right chapter for this problem. Nobody connected them.

The coordinate reframing in [turn_ndo_ch02.tex](#) is a real improvement — “not a threshold an agent crosses but a point it occupies” — because it stops consciousness being a club with an entrance exam. But it does not fix the underlying problem, which is that a position in a lattice of computational strength is still a statement about capacity, and capacity is not experience. It relocates the assertion.

What would fix it, in your own terms: consciousness is not the resolution an agent has, it is the agent’s *model of its own resolution* — the second-order structure by which a system represents which distinctions it is currently making and could make. That is buildable here: the modal depth of the formulae a learner is currently funding is a quantity the learner could itself have a formula about, and a system that tracks that quantity has something extremely like attention.

Sentience is a budget line, not an affect

[turn_hypercomp_ch04.tex](#) makes ζ a distinguished component of the vectorial account, driven by a reinforcement signal on predictive performance, gating what the agent can afford. The remark that “an agent that feels bad literally cannot think as well” and the ensuing vicious cycle are the strongest part of the chapter, and the structural parallel to depression is worth having.

But as constructed, ζ is indistinguishable from being poor. Four things affect is and ζ is not:

1. *It is multi-dimensional.* Valence and arousal are not one number, and the difference mat-

ters functionally: high-arousal negative and low-arousal negative states produce opposite policies. You already have a *vector* $\vec{A} = (A_1, \dots, A_k)$ and you singled out one component. Why not a projection of the whole vector, or a second-order signal about its rate of change?

2. *It is intentional.* Affect is *about* something — fear of that, disgust at this. ζ is about the agent’s aggregate track record. The framework has the apparatus for aboutness: a feeling could be indexed by the namespace whose predictions are failing.
3. *It modulates policy, not just budget.* Fear does not make you poorer; it makes you prefer different actions at the same wealth. In the book, everything affect does is done by making experiments unaffordable, which conflates a mood with a liquidity crisis.
4. *It has set points and homeostasis.* ζ has a threshold and a drift and no target. Motivational systems have regulated variables.

And the deeper problem: **there is exactly one drive**. “The motive is metabolic” ([sci:S1.3](#), R4) is elegant, and it solves the scorekeeper problem honestly, which most frameworks do not. But a mind with one drive is a bacterium. Real cognitive systems have competing, incommensurable drives and the interesting work is in the commensuration.

You have the machinery for this and it is in the wrong chapter. [ch:eng](#) gives multiple token colours, engines that convert between them, a conversion graph whose cycles are an error-correcting code, and — the key piece — the virtual token, which exists exactly when no cycle of conversions pays for itself. **That is a theory of commensurable drives, and the case where arbitrage is present is a theory of incommensurable ones.** Rename nothing, add one section: the token colours are drives, the engines are the trade-offs an animal makes between them, and incommensurability ([turn_rholife_ch04.tex](#) §“Incommensurability”) is what a value conflict is.

No attention, and the framework is shaped like one

A bounded learner with a tree-shaped hypothesis space, a metered search, and a wealth-bounded branching factor ([sci:prop:branching](#)) is a system that *must* allocate, and [§sci:sec:search](#) declines to say how. The book is explicit about this being deliberate.

I understand the reason — you want to show which features are forced, and policy is one of three genuinely free parameters. But the consequence is that a part titled *Ecology as Cognitive Architecture* contains no attention mechanism, and attention is the first thing anyone building a cognitive architecture has to supply.

What I would want, and what I think falls out: attention is the allocation of the metered search budget across the branches of [sci:prop:branching](#), and the *constraints* you have derived are already strong enough to characterise the space of admissible attention policies even without picking one. State that as a result. “Here is the space of possible attention mechanisms for a mortal learner, and here is what every member of it must satisfy” is a stronger contribution than picking one, and it is what you already have.

The unity problem, which the book almost names

What makes the four skill-ecologies of `scm:sec:composite` one mind rather than four minds sharing a pod? `ch:eng` answers: closure. A region whose internal media are closed and whose boundary media are not is an individual, subject to an isoperimetric condition.

That is an *economic* criterion — who can be named from outside, who pays for what. And the book knows it is not enough: `turn_ndo_ch02.tex` §“Choice and the drawing of boundaries” distinguishes two criteria for a boundary, economic and semantic, and says neither implies the other. **The semantic criterion is never supplied.** So the framework has a rigorous account of what makes something an economic individual and no account of what makes something a single subject.

This is the unity of consciousness, and the book should name it as such. My guess at where the answer lives, in your terms: a region is one subject when its self-model is one model — when the fixed-point solution of `eng:S8` that the region *represents itself as* is a single generator rather than several. That makes unity a property of the self-model rather than of the plumbing, which is where I think it belongs, and it makes split-brain and dissociative phenomena the case where the plumbing and the self-model disagree. It is also checkable in your framework, because generators are formulae and you can ask how many there are.

One thing the book gets right that almost nobody does

The claim that the true theory of an environment can be dominated by a cruder theory that costs a fraction as much and wins nearly as often (`sci:rem:yieldbias`, and demonstrated with numbers in `turn_rholife_ch02.tex`) is correct, important, and rarely stated. Most accounts of rationality treat approximation as a regrettable concession; this treats it as the equilibrium.

That result is also, I would argue, the real answer to why minds have the character they do — confabulating, coarse-graining, committed to serviceable fictions — and it is stated as an ecological observation rather than as a thesis about cognition. It is a thesis about cognition. Promote it.

6 Where the five disagree

A synthesis that averaged these reports would produce mush. The disagreements are the useful part, so here they are, adjudicated.

Rigour versus legibility (Spivak vs. Jaimungal)

Spivak wants the coherence data exhibited: distributive laws, cartesian lifts, triangle identities. Jaimungal wants the book to be holdable by a smart non-specialist and thinks it is already seventy pages too slow at the front.

These are not actually opposed, because they concern different chapters. Spivak's demands land on Part I and the fibration; Jaimungal's on the ordering and framing of the whole. **Resolution: do the category theory properly and shorten its presentation.** A correctly stated adjunction is briefer than a hand-waved one, because the hand-waving is where the paragraphs of apology go.

Buildability versus falsifiability (Goertzel vs. Tegmark)

Goertzel wants an architecture that runs; Tegmark wants a number that could come out wrong. Under a fixed writing budget these compete directly, and the book cannot do both well in this draft.

Resolution: Tegmark wins for The Turn, Goertzel wins for the next paper. The reason is that the falsifiable content already exists and is under-presented (the overcounting bias, the depth-before-population prediction, the $\log \text{CN}/h$ ratio), whereas the architecture content would be new writing of a kind that would double the ecology part. Take the free win now. But adopt Goertzel's two cheapest asks, because they cost paragraphs rather than chapters: state that inference control in this framework is a choice-type discipline, and repair the instrument-building objection to `thm:ai-bound`.

Whether the consciousness chapters are salvageable (Bach vs. everyone)

Tegmark, Spivak and Jaimungal all treat chapters 48–50 (`turn_hypercomp_ch04–ch06`) as chapters with fixable defects: restore the coordinate revision, drop the retracted amplitudes, decide the lattice. Bach says the defect is not fixable by restoration, because “consciousness is a position in a lattice of computational strength” is a claim about capacity and no amount of tidying makes capacity into experience.

Bach is right about the diagnosis and the other three are right about the sequencing. Restore first — the contradiction is indefensible and cheap to fix — and then write the section Bach is asking for, which is not a defence of the oracle definition but a different claim standing beside it: that what a system reports as phenomenal is the shortfall between its self-model and itself. That claim is derivable from machinery already in the book and does not require the oracle definition to be true.

Whether to keep the ordinal tower at all

You decided earlier that the tower stays, as a way of introducing the intuitions, with the lattice as the real structure. Spivak and Bach both push back, for different reasons: Spivak because the tower as presented is not a fibration and its repair is a different object; Bach because a chain-shaped shadow of a lattice invites exactly the threshold reading the book is trying to escape.

Resolution: keep it, but demote it explicitly and early. The tower should be introduced as an idealisation with a stated defect — one sentence at first use, not a retraction three chapters later. [turn_hypercomp_ch02](#)'s missing closing section was going to do exactly this.

Assembly Theory: friendly amendment or takeover?

Tegmark reads Part VI as a friendly amendment to Cronin and Walker that the book under-sells. Jaimungal reads it as a takeover the book has not earned, because the answers are internal to the rho calculus and not operational for a chemist.

Both are right about different halves. The *copy-number* work is a friendly amendment and should be presented as one, loudly. The *assembly-index* identification is a takeover and is currently asserted as a proposition without proof. **Split the two and give them different rhetorical registers.**

7 Cross-cutting findings

Six items that more than one reviewer hit, ordered by how cheap they are to fix relative to how much damage they do.

1. Four of five revisions listed in INTEGRATION-NOTES-2.md were never committed

draft3/INTEGRATION-NOTES-2.md records five revised files. Only `turn_hypercomp_ch07.tex` is present in the repository. Missing:

File	What the notes say it contained
<code>turn_hypercomp_ch01</code>	consciousness as a coordinate; organisation paragraph
<code>turn_hypercomp_ch02</code>	new closing section, <i>Two Things This Construction Gets Wrong</i>
<code>turn_hypercomp_ch05</code>	the ordinal remark and the comparison table
<code>turn_rholife_ch04</code>	forward pointer into ch. 39

Verified against the built tree: `turn_hypercomp_ch01.tex` contains zero occurrences of “coordinate” and still reads “*Consciousness* is an ordinal-valued threshold property”; `turn_hypercomp_ch05.tex` still has `remark[Consciousness is Ordinal-Valued]` and the table row “Changes: By threshold (have it or not)”; `turn_rholife_ch04.tex` contains zero references to `ch:apr` or `ch:agy`.

The visible damage is a forward reference that lies. `turn_ndo_ch02.tex` tells the reader that consciousness in the chapters that follow is “not a threshold an agent crosses but a point it occupies, with the ordinal tower... the chain-shaped shadow of the lattice that point really lives in.” Two chapters later the book says consciousness is discrete, ordinal, and changes “by threshold (have it or not).” `turn_conclusion_ch01.tex` sides with the older view (“the first continuous, the second not”).

This is the highest-value repair in the document and it is a restoration, not new work.

2. Retracted apparatus still load-bearing in two chapters

`turn_causality_ch09.tex` withdraws the complex-amplitude reading. Still using it:

- `turn_sciencebot_ch05.tex`, $\text{Message Fidelity} = |W(\gamma)|^2$ — the foundation of all of Part IV.
- `turn_hypercomp_ch05.tex`, ϵ -approximate oracle access via the path integral.
- `sec:hyper-gaps` Gap 7, which asks for the oracle amplitude to be made precise via the quantum Gillespie algorithm.

Neither use needs amplitudes; the stochastic instance supplies a multiplicatively degrading fidelity and a probability, and the book says that instance is unobstructed and is what the implementation runs.

3. `thm:compact-consensus` is unproved as written

`turn_sciencebot_ch06.tex`. The family $\{\llbracket \varphi_n \rrbracket\}$ is decreasing, so $\bigcup_{n \leq N} \llbracket \varphi_n \rrbracket = \llbracket \varphi_1 \rrbracket$ and the agents claimed to be uncovered are covered. The Conclusion calls this the sharpest result of its part.

4. Two objects named “the fibration”

$\Phi : \mathbf{GSLT}_{\text{TC}} \times \text{Ord} \rightarrow \mathbf{GSLT}$ (`sec:fibration`) is an indexed family with contradictory variance, asserted as built. $p : \text{Hyp} \rightarrow \mathbf{GSLT}$ (`turn_ndo_ch02.tex`, open problem (ii)) is the right object, flagged as unbuilt. The reader meets the assertion first.

5. `turn_hypercomp_ch08.tex` is stale

Describes the material as “the three papers in this series”; credits the ordinal treatment of consciousness as “novel” after two later chapters argue it is wrong; disposes of Friston in three sentences and Tononi in two.

6. The falsifiable content is real and is filed as remarks

Every reviewer noticed some version of this. The claims that could come out wrong are `scm:prop:geom` with `scm:rmk:falsepos` (coarse instruments inflate copy number exponentially, biased toward the biosignature), `scm:rmk:biosig` (the ratio $\log \text{CN}/h$ should cluster near $\log(1/p)$ for architectures in use), and `scm:rmk:shed` (lineages under scarcity shed depth before population). Three remarks, in one chapter, near the end of a 570-page book that says four times that it is stated precisely enough to be wrong.

8 Revision plan

Four tiers. Tier 0 is restoration and repair of things the book already believes; Tier 1 is structural moves that change what the reader encounters and in what order; Tier 2 is new writing that the reviewers converged on wanting; Tier 3 is what to leave alone.

Effort is in units of a working session. “Files” names what changes.

Tier 0 — Repairs. Do these first; none needs new research.

	Action	Files	Effort
R1	Restore the four lost <code>INTEGRATION-NOTES-2</code> revisions: the coordinate framing in ch01, <i>Two Things This Construction Gets Wrong</i> in ch02, the ordinal remark and comparison table in ch05, the forward pointer in rholife ch04. Then fix the Conclusion’s “the first continuous, the second not”.	<code>turn_hypercomp_ch01/02/05</code> , <code>turn_rholife_ch04</code> , <code>turn_conclusion_ch01</code>	1–2
R2	Rewrite message fidelity in the stochastic semiring; rewrite ϵ -approximate oracle access as a probability rather than a squared amplitude; delete or restate Gap 7.	<code>turn_sciencebot_ch05</code> , <code>turn_hypercomp_ch05</code> , <code>turn_hypercomp_ch07</code>	1
R3	Repair or withdraw <code>thm:compact-consensus</code> . The likely fix is to index the formulae by “fewer than n ” and take an increasing cover. If it does not go through, demote to a conjecture and amend the Conclusion, which currently calls it the part’s sharpest result.	<code>turn_sciencebot_ch06</code> , <code>turn_conclusion_ch01</code>	1
R4	Decide Scott or Stone. With negation in HML the space is Stone; say so, and connect to the ultrametric hypothesis space of <code>sci:sec:lattice</code> .	<code>turn_sciencebot_ch06</code>	0.5
R5	Downgrade Φ from “fibration” to “indexed family”, name the intended variance, and add a forward pointer to <code>turn_ndo_ch02</code> open problem (ii) as the object that would replace it.	<code>turn_hypercomp_ch02</code>	0.5
R6	Rewrite <code>turn_hypercomp_ch08</code> from scratch, or delete it and fold one paragraph into <code>turn_conclusion_ch03</code> .	<code>turn_hypercomp_ch08</code>	0.5
R7	Resolve the d^* / “separates at” ambiguity in the worked composite; either justify the identification in one sentence or recompute the floor at 24 and update the two percentages.	<code>turn_origins_ch09</code>	0.5
R8	Delete the numeral 75 from <code>scm:rmk:groupsize</code> , keep the scaling law. Fix “quarks” and drop 10^{80} in favour of a scale-free statement of the diameter/coherence condition.	<code>turn_origins_ch09</code> , <code>turn_sciencebot_ch05</code>	0.5

Tier 1 — Structural. These change the reader’s path.

S1. Promote the falsifiable content into its own chapter. Take `scm:prop:geom`, `scm:rmk:falsepos`, `scm:rmk:biosig` and `scm:rmk:shed` out of `turn_origins_ch09` and make a short chapter — *What Would Show This Is Wrong* — at the end of Part VI, with the overcounting result as its lead. Add the Fermi estimate of k for tandem MS. This is the single highest-leverage editorial move available and it costs about two sessions.

S2. Extend the status column to every part. `turn_causality_ch10`’s CON/THM/CNJ table is the best editorial device in the book and appears in exactly one part. Part V and Part VI need it more than Part III does, because their claims are bolder. Then delete three of the four occurrences of “precisely enough to be wrong” — the column does that work now.

S3. Give Part II a two-page on-ramp. “What you need from Part I”: the four or five constructions the ecology chapters actually use, with pointers. This cashes the Overarching Introduction’s promise that a reader may begin at Part II, which nothing currently supports.

S4. Move the finiteness confrontation to the front of Part V. `turn_entropy_ch01`’s closing admission — that if every physical aperture carries only finite contention the framework predicts no physical hypercomputation — should open the part, stated in its strongest form, with what survives stated immediately after. What survives is the location claim, which is the part’s real contribution and is independent of whether the tower is inhabited.

S5. Promote crossover. `sci:sec:repro` is a subsection carrying the book’s most novel mechanism: crossover on quoted code with contexts as loci and a type discipline available from OSLF. Make it a chapter of Part II. Cite the genetic-programming literature it is implicitly contradicting.

S6. Split the two Assembly Theory claims. Present the copy-number work as a friendly amendment (it is, and it is good). Downgrade `prop:assembly-index` from a proposition to a proposed identification, and state the missing object — a functor from chemical reaction systems to ciGSLTs preserving minimum path length, with AT’s reuse discipline — as the central open problem of Part VI.

Tier 2 — New writing the reviewers converged on

N1. The self-model section. (*Bach; endorsed in effect by Jaimungal, who wants the pivot inside The Turn.*) The germ is a quotation of the learner’s own code; the hypothesis language is a logic of namespaces; the learner’s namespace is a namespace. Therefore a learner can form hypotheses about itself, priced by the same assay. Its self-model is necessarily coarser than itself, the coarsening is not represented, and it reports a self simpler and more unified than it is. Place at the end of Part II or as the closing section of `ch:eng`. Retire Gap 8 by replacing it with this.

N2. Drives, in the engine chapter. (*Bach.*) Token colours are drives; engines are the trade-offs; the virtual token is what makes drives commensurable; arbitrage is what makes them incommensurable. One section, no new mathematics, and it turns `ch:eng` from an economics chapter into a motivational-systems chapter.

N3. The two-lattices paragraph. (*Tegmark.*) Name AC-strength and Weihrauch degree as different structures, say which one Part V uses, and state the relation between them as an open problem rather than an identification. Half a page in `turn_entropy_ch01` or `turn_ndo_ch01`.

N4. The comparison table. (*Jaimungal.*) One page, early in The Turn: IIT, FEP, Assembly Theory, ruliad, global workspace, and where this book disagrees with each. Every row is already argued somewhere; none is collected.

N5. The interchange lattice. (*Spivak.*) `rmk:two-erasures` says the cost and history counits do not factor through each other and that the failure is graded. Stated properly in Part I, that is the observation that no distributive law exists in either direction and that what exists instead is a lattice of lax interchange maps. It also makes the entropy chapter’s “supply of positions” rest on a proposition.

N6. Instruments. (*Goertzel.*) One paragraph qualifying `thm:ai-bound`: the bound is on an ecology resolving unaided, and instrument-building is precisely how minds evade it. The bound then applies to ecology-plus-instruments, which is a larger object with a larger index. Without this the slogan’s natural reading is false.

N7. Inference control is a choice-type discipline. (*Goertzel.*) One paragraph connecting `sci:sec:search` to `ch:chs`. Ties Part II to Part V across a gap neither part currently bridges.

Tier 3 — Leave alone

- `turn_gslt_ch05`, *The Category of GSLTs*. Spivak’s only unqualified praise. Do not touch it.
- `turn_causality_ch09`. The retraction is a model of how to withdraw a claim. The problem is downstream, not here.
- `turn_rholife_ch02`, the worked game. It is the only place a claim is checked against a number and the numbers hold up.
- The decision to keep the ordinal tower as intuition-scaffolding. Two reviewers push back; the answer is to demote it explicitly (R1, R5), not to remove it.
- The Pledge. Out of scope, and left untouched by design.

Suggested order

Tier 0 entire, in one pass, since R1–R3 are the ones that make the book self-contradictory and they are all restorations or local repairs. Then S1 and S2, which are the cheapest way to change how the book reads to a sceptic. Then N1, which is the one piece of new writing every reviewer wanted in some form and which puts the pivot inside The Turn where Jaimungal argues it belongs. Everything else can wait for the next draft.

The five voices are simulated readings, constructed to keep distinct sets of commitments in play. Attribute the criticism to the exercise, not to the people named.